

通訊網路實驗

Android & Python Programming
Android Studio

Dept. of Electrical and Computer Engineering (ECE)
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評分標準 & 注意事項

- 出席 30%
- Demo 30%
- 結報 40%
 - e3上有學習單
 - 檔名：學號_姓名_Labx.pdf
 - 交pdf到對應的資料夾中 (期限一週)

課程大綱

- Android Studio Intent & Bundle
- Android Studio 猜數字小程序

Demo項目

- Q1: 換頁
 - Q2: 猜數字小程序
 - Q3: Intent&Bundle
- Q1、Q2、Q3皆寫在同一個專案內，開專案時language記得選Java

Intent(意圖)

- 在activity之間轉換時使用
- 可以用來傳遞資料
- 使用.putExtra()加入資料到intent內， getIntent()接收intent
- 主要在傳值

Bundle

- 在不同activity之間傳遞資料
- 需要Intent協助才能完成資料傳遞
- 使用.putExtras(bundle)傳送bundle，.getExtras()接收bundle
- 可以傳遞複數種資料
- 主要在傳資料
 - ❑ 傳String的資料時要用 .putString()
 - ❑ 傳Int的資料時要用 .putInt()

假設從頁面A傳遞到B，再傳遞到C (Intent)

A頁面中:

```
Intent intent = new Intent();
intent.setClass( packageContext: MainActivity.this, MainActivity2.class);
intent.putExtra( name: "String", value: "MainActivity中的值");
intent.putExtra( name: "int", value: 11);
startActivity(intent);
```

B頁面中:

需要先在B頁面中接收資料

```
Intent intent = getIntent();
String string = intent.getStringExtra( name: "String");
int key = intent.getIntExtra( name: "int", defaultValue: 0);
```

然後再發資料到C頁面

```
Intent intent = new Intent();
intent.setClass( packageContext: MainActivity2.this, MainActivity3.class);
intent.putExtra( name: "String1", string);
intent.putExtra( name: "int1", key);
intent.putExtra( name: "boolean", value: true);
startActivity(intent);
```

假設從頁面A傳遞到B，再傳遞到C (Bundle)

A頁面中:

```
Intent intent = new Intent();  
intent.setClass( packageContext: MainActivity.this, MainActivity2.class);  
Bundle bundle = new Bundle();  
bundle.putString("String", "MainActivity中的值");  
bundle.putInt("int", 11);  
intent.putExtras(bundle);  
startActivity(intent);
```

在B頁面接收資料:

```
Bundle bundle = this.getIntent().getExtras();
```

然後在B頁面中傳送資料:

```
Intent intent = new Intent();  
intent.setClass( packageContext: MainActivity2.this, MainActivity3.class);  
//可以傳給MainActivity3額外的值  
bundle.putBoolean("boolean", true);  
intent.putExtras(bundle);  
startActivity(intent);
```



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建立一個空的activity

The screenshot illustrates the process of creating a new empty activity in Android Studio. The interface is divided into three main sections:

- Left Panel (Project Explorer):** The 'res' folder is expanded, and the 'layout' sub-folder is selected. A red box highlights the 'New' button in the context menu.
- Center Panel (Code Editor):** The 'New' context menu is open, showing various options. The 'Activity' option is highlighted with a red box. A sub-menu is also visible, showing 'Empty Views Activity' as the selected option, also highlighted with a red box.
- Right Panel (New Android Activity Dialog):** The 'New Android Activity' dialog is open. It shows the 'Empty Activity' template. The 'Activity Name' is set to 'MainActivity2', the 'Layout Name' is 'activity_main2', and the 'Package name' is 'com.example.lab2test'. The 'Source Language' is set to 'Java'. A red box highlights the 'Finish' button at the bottom right.

所有的Name都可以自行變更



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Q1

Q1

- 先設計出App一打開時顯示的主頁面，然後繼續做換頁設定
- 步驟：
 - ❑ 開啟一個空的新專案
 - ❑ 在xml裡面(如右圖)
 - 新增兩個EditText，一個輸入學號，另一個輸入名字
 - 新增一個Button用來轉到下一個activity

學號

Name

START GAME

Q1

- 在MainActivity.java中依序做下列

- ❑ 宣告EditText、Button等
- ❑ 使用findViewById連結xml中的元件

```
xxx = (Button) findViewById(R.id.xxx);
```

- ❑ 藉由.setOnClickListener函數完成按下button後需要做的動作
- ❑ 接著建立一個空的activity，建好之後先不用動他

Q1

□ 接上頁，在MainActivity.java中

□ 放在.setOnClickListener裡面

- Intent intent = new Intent();
intent.setClass(MainActivity.this, MainActivity2.class);
- Bundle bundle = new Bundle();
bundle.putString("變數名稱", 變數的內容);
- intent.putExtras(bundle);
- startActivity(intent);

```
nextbutton.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        Intent intent = new Intent();  
        intent.setClass(packageContext MainActivity.this, MainActivity2.class);  
        Bundle bundle = new Bundle();  
        bundle.putString("變數名稱", 變數內容);  
        intent.putExtras(bundle);  
        startActivity(intent);  
    }  
});
```



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Q2

Q2

- 猜數字小程序
- 規則：
 - ❑ 在1~50裡面猜一個數字
 - ❑ 猜測數字後會得到新的提示(紅框框內顯示的文字)
 - ❑ 每猜一次猜測次數+1
 - ❑ 猜對後顯示猜對且會有最佳紀錄
 - ❑ 按下重新開始後可以進行下一輪遊戲



The screenshot shows a web-based number guessing game interface. At the top left is a green button labeled "重新開始" (Restart). In the center, the title "猜數字" (Guess Number) is displayed. Below the title is a red rectangular box containing the prompt "請輸入1~50的數字" (Please enter a number between 1 and 50). Underneath this is a text input field. Below the input field is a dark grey button labeled "送出" (Submit). At the bottom, there are two status indicators: "猜測次數：0" (Guessing count: 0) and "最佳紀錄：0" (Best record: 0).

Q2

- 將Q2的程式放在第二個activity中
 - Hint:
 - 在Activity_main2.xml中
 - 新增一個EditText，用來輸入猜測的數字
 - 新增一個Button用來送出猜測的數字
 - 新增三個TextView用來記錄提示、猜測次數、最佳紀錄
 - 新增一個Button用於重新開始遊戲
- (如前頁圖示)

Q2

□ 在MainActivity2.java中

❑ 宣告EditText、Button、TextView

❑ 宣告變數

```
int randNum=0, history=999;  
int min=1, max=50, count=0;
```

❑ 藉由findViewById連結xml中的元件

❑ 設定randNum=(int) (Math.random() * 50 + 1);

❑ 按下”送出”按鈕後的判斷式(見下頁)

❑ 設定按下”重新開始”按鈕的功能

e.g.重新選擇randNum、min/max和提示改為初始值、猜測次數歸零

Q2

□ Hint:

❓ 按下“送出”按鈕後的判斷式

```
outputbutton.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        int in = Integer.parseInt(input.getText().toString());  
        if (in <= max && in >= min) {  
            if (in > randNum) {  
                max = in;  
                //  
            } else if (in < randNum) {  
                min = in;  
                //  
            } else if (in == randNum) {  
                if (count < history) {  
                    //  
                }  
                //  
            }  
        } else {  
            提示.setText("請輸入" + min + "~" + max + "的數字, 請輸入正常值");  
        }  
    }  
});
```



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Q3

接收bundle的資料

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main2);  
  
    Bundle bundle = this getIntent().getExtras();  
    String idnum = (String)bundle.getString( key: "idnum");  
}
```

- `Bundle bundle = this.getIntent().getExtras();`
 String “變數名稱” = (String)bundle.getString(“在bundle內的變數名稱”);
- 接收上個activity的資料傳遞時，要放在onCreate 後才可以
 不然還沒初始化就接收app會閃退

Q3

- 在剛剛Q2 `super.onCreate(savedInstanceState);`
`setContentView(R.layout.activity_main2);` 的下面接收Q1傳送的資料
- 在剛剛Q2的介面，新增一個Button用來轉到下一個activity
- 步驟：
 - ❑ 宣告新的Button、findViewById
 - ❑ 新建立一個空的activity



Q3

- 在MainActivity2.java中
 - ▣ 剛剛新設置的button的setOnClickListener裡面要把Q1的學號、姓名以及Q2得到的最佳紀錄一起傳到新的activity
 - ▣ Hint:要傳數字可以用bundle.putInt(“變數名稱”, 變數的內容);

Q3

- 加入back button，在按下之後回到主頁面

BACK

學號: 0810XXX

姓名: Bob

最佳紀錄: 4

可能遇到的問題

```
Build test: failed At 2023/11/16 下午 02:57 with 2 errors 15 sec, 652 ms
  Download info
  :app:checkDebugAarMetadata 1 error 1 sec, 76 ms
    An issue was found when checking AAR metadata:
    An issue was found when checking AAR metadata:

An issue was found when checking AAR metadata:

1. Dependency 'androidx.activity:activity:1.8.0' requires libraries and applications that
   depend on it to compile against version 34 or later of the
   Android APIs.

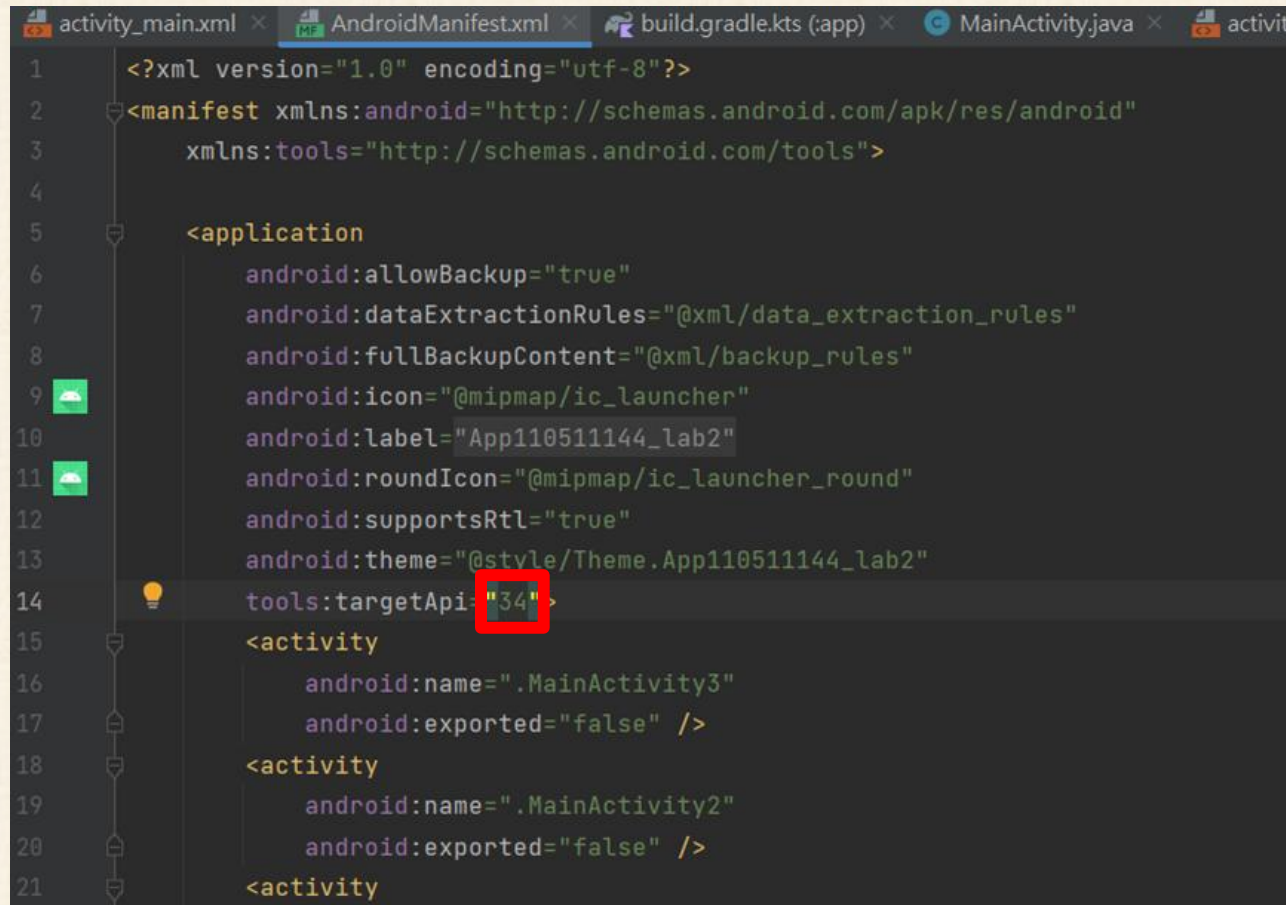
   :app is currently compiled against android-33.

Recommended action: Update this project to use a newer compileSdk
of at least 34, for example 34.

Note that updating a library or application's compileSdk (which
allows newer APIs to be used) can be done separately from updating
targetSdk (which opts the app in to new runtime behavior) and
minSdk (which determines which devices the app can be installed
on).
```

解決方法

(1)在AndroidManifest.xml中，將targetApi改成34



```
1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3          xmlns:tools="http://schemas.android.com/tools">
4
5      <application
6          android:allowBackup="true"
7          android:dataExtractionRules="@xml/data_extraction_rules"
8          android:fullBackupContent="@xml/backup_rules"
9          android:icon="@mipmap/ic_launcher"
10         android:label="@string/app_name"
11         android:roundIcon="@mipmap/ic_launcher_round"
12         android:supportsRtl="true"
13         android:theme="@style/Theme.App110511144_lab2"
14         tools:targetApi="34">
15
16         <activity
17             android:name=".MainActivity3"
18             android:exported="false" />
19
20         <activity
21             android:name=".MainActivity2"
22             android:exported="false" />
23
24         <activity
```

□ (2)將build.gradle.kts(:app)中的compileSdk跟targetSdk改成34

